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Micro-aeration and leachate recirculation for the acceleration of landfill stabilization: Enhanced hydrolytic acidification by facultative bacteria

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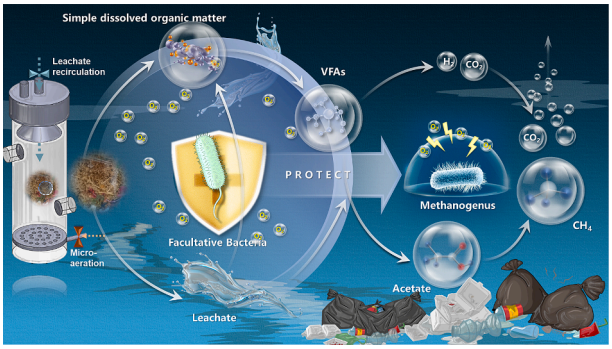
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HIGHLIGHTS

- Micro-aeration and leachate recirculation accelerate the stabilization of landfill.
- The integrated process promotes the enrichment of facultative bacteria.
- Bacteria that cause hydrolytic acidity benefit the stabilizing process.
- Synergy between methanogens and acidifying bacteria is enhanced in integrated process.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

The long duration of landfill stabilization is one of the challenges faced by municipalities. In this paper, a combination of micro-aeration and leachate recirculation is used to achieve rapid degradation of organic matter in landfill waste. The results showed that the content of volatile fatty acids (VFAs) in the hydrolysis phase increased significantly and could enter the methanogenic phase quickly. Until the end of the landfill, the removal rates of chemical oxygen demand (COD), total phosphorus (TP) and ammonia nitrogen (NH_4^+–N) by micro-aeration and leachate recirculation reached 80.17 %, 48.30 % and 48.56 %, respectively, and the organic matter degradation rate reached 50 %. Micro-aeration and leachate recirculation enhanced the abundance of facultative hydrolytic bacteria such as *Rummeliibacillus* and *Bacillus* and the oxygen tolerance of *Methanobrevibacter* and *Methanoculleus*. Micro-aeration and leachate recirculation improved the organic matter degradation efficiency of landfill waste by promoting the growth of functional microorganisms.

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1. Introduction

As the final stage of comprehensive waste treatment and disposal, landfill remain the dominant technology in developing nations (Yu et al., 2022). Although the number of landfills in developed countries (e.g., the United States and Japan) had declined since the 1990 s, landfills in developing countries continue to increase, and China was expected to be the largest landfill country by 2030. As of 2021, China had 644 landfills that disposed of 77.7 million tons of waste, and landfills constitute 45.6 % of environmentally sound treatment plants (NBSC (National Bureau of Statistics of China), 2021). Anaerobic landfills do not promote microbial growth, resulting in a reduction in the biodegradation rate of organic wastes (Viegas et al., 2021). Many landfills stabilize only after decades or centuries of decomposition (Lee et al., 2022). Landfills occupy a large area, rendering land resources scarcer and severely impeding the development of urbanization. In addition to occupying a considerable amount of land, landfills also severely limit the development of urbanization (Huang et al., 2022). In addition to reducing the risk of environmental pollution, the large amount of stabilized land can be used for commercial or residential purposes (Zhang et al., 2023).

Therefore, it is crucial to expedite the stabilization process of landfills. As a solution, leachate recirculation could significantly reduce the amount of leachate to be treated, increase the water content and redistribute nutrients and microorganisms in the landfill layer (Xu & Zhang, 2022). The reinjection of leachate into the interior of the landfill significantly accelerates waste decomposition and stabilization, increases landfill gas production, and rapidly produces settling. Tong et al., (2015) demonstrated the effect of leachate recirculation on the hydrolysis of food waste in a leaching bed reactor, finding that it increased the possibility that hydrolytic bacteria would come into touch with the solid surface. In general, hydrolytic acidification are considered rate-limiting steps in the degradation of large particulate organic wastes (Vavilin et al., 2010). Several strategies to improve solid-phase solubility have been reported, where leachate recirculation has been identified as one of the key parameters to improve the efficiency of hydrolysis and acid production, which helped to improve mass transfer and redistribute enzymes and microorganisms (Zhang et al., 2022a, Zhang et al., 2022b; Gardner et al., 1993).

However, excessive leachate recirculation could lead to organic load overload and acid accumulation, impeding methane production, especially for anaerobic bioreactors (Grossule & Lavagnolo, 2020). Stabnikova et al., (2008) demonstrated the ability of leachate recirculation to extract organic substrates from food waste. However, high concentrations of metabolic intermediates in leachate tended to inhibit and retard the hydrolysis process. Wang et al., (2021) reported that the accumulation of volatile fatty acids (VFAs) affects the activity of certain specific enzymes and bacteria, thereby inhibiting hydrolysis and acid production. Long-term leachate recirculation can also lead to ammonia nitrogen accumulation in landfills, inhibiting the degradation of landfill waste (Hossain et al., 2009). Previous studies had found that aerobic aeration to the interior of landfill piles could reduce this inhibition and improve solid waste solubilization and hydrolysis in landfill simulator reactors (LSRs) (Ike, 2009; Pinjing, 2011). However, the conversion of the entire landfill to an aerobic condition incurs a considerable energy cost, rendering it an uneconomical treatment option for practical engineering purposes.

In recent years, micro-aeration technology has been maturely applied in organic wastewater treatment, which can effectively remove organic pollutants in wastewater (Yang et al., 2023; Hu et al., 2012). (Cottingham et al., 1999) showed that micro-aeration could promote nitrification in artificial wetlands, but the removal of total nitrogen (TN) was limited by the insufficient carbon source. (Ouellet-Plamondon et al., 2006) conducted micro-aeration at the inlet end of a horizontal flow wetland planted with calabash, and the experimental results showed high removal of chemical oxygen demand (COD) and TN. The removal rate of H_2S reached more than 90 % when domestic wastewater was aerated in an upflow anaerobic sludge bed (UASB) (Castro et al., 2023). 2, 4-dinitrophenol is a raw material for synthetic compounds such as pesticides, pharmaceuticals and dyes, and widely exists in industrial wastewater. Under the condition of micro-aeration, a large amount of oxidoreductase is produced, which leads to the degradation of 2, 4-dinitrophenol (Zhang et al., 2022a, Zhang et al., 2022b). Micro-aeration can improve the activity of microbial hydrolytic enzymes and the removal rate of organic matter in the aeration tank in the wastewater treatment system, and can enhance the synergistic effect of hydrolyzing bacteria and acid-producing bacteria. In addition, micro-aeration results in higher diversity and abundance of hydrolyzing bacteria, which improves the hydrolysis of complex organic matter and enhances biodegradability (Zhang et al., 2022a, Zhang et al., 2022b).

In addition, the positive effect of microaeration on the production and consumption of VFAs has been progressively demonstrated (Liu et al., 2023; Shang et al., 2023). Yu et al., (2020) observed that appropriate oxygen increased the rate of hydrolytic acidification during blackwater digestion and the abundance of colleague hydrotropic methanogenic bacteria was also enriched in this environment. Nguyen et al., (2019) discovered that the content of volatile fatty acids was reduced in a short period of time by 67.4 %, which promoted the recovery of methanogenic performance and enhanced overall system stability. According to research, micro-aeration can drive facultative anaerobic bacteria to release more extracellular enzymes to improve lignocellulosic biomass hydrolysis, hence increasing methane generation potential (Zheng et al., 2019). Micro-aeration in anaerobic digestion and wastewater treatment processes has received increasing attention due to its potential economic and environmental benefits (Cao et al., 2022). However, there are few studies on the use of micro-aeration in landfills to expedite the decomposition of VFAs accumulated during landfilling, and additional research is required.

Although the positive effect of micro-aeration on acid accumulation can be expected, it is currently unclear how the combined air–water regulation of micro-aeration and leachate recirculation accelerates the stabilization of landfill waste. This paper can provide a scientific reference for establishing a study on rapid landfill stabilization.

The recirculation of pretreated leachate allows the neutralizing substances of leachate to diffuse throughout the landfill, while micro-aeration enables facultative bacteria within the landfill to function, and the VFAs accumulated can be rapidly utilized to accelerate the stability of the landfill. Therefore, the objectives of this study were: (i) to evaluate the effects of leachate recirculation and micro-aeration on the landfill stabilization process by monitoring leachate, gas and solids indicators and comparing with conventional anaerobic landfills; (ii) to investigate the degradation effects of micro-aeration and recirculation systems on accumulated VFAs; and (iii) to investigate the bacterial and archaeal community structure of landfill using microbiological high-throughput sequencing mechanism of landfill stabilization under combined gas–water conditioning. The results of this study can serve as a technical reference for the development of landfill process upgrades and contribute to society's sustainable development.

2. Materials and methods

2.1. Feedstock

The landfill simulation experiments were obtained from domestic waste from the dormitory area of the university campus and leftover canteen refuse from the cafeteria. The waste was screened and pulverized before landfilling, and the proportions were adjusted according to the composition of MSW before being deposited into the landfill simulator. The composition ratios of landfill waste in this simulation experiment are shown in Table 1.

Table 1
The composition of waste inside the landfill tanks.

Waste type	Ratio (%)
Food waste	55 %
paper	15 %
plastic	10 %
Branch (lignin)	10 %
Leaves (cellulose)	10 %

2.2. Design and operation of the experimental device

The landfill simulation experiment was conducted in 4 landfill simulation reactors (LSRs) made of organic glass, as shown in Fig. 1. The inner diameter of the LSR is 400 mm and the height is 800 cm. The top of the LSR is equipped with an external leachate recirculation pipe and gas discharge pipe, which are connected to an external wet gas flow meter to facilitate the collection and measurement of the generated landfill waste. Two sampling ports are set on the side of the column, upper and lower, to facilitate uniform sampling. At a height of 10 cm from the bottom of the column, a perforated plate with uniform openings is installed, which also serves as a support layer. A leachate collection chamber is set up below the perforated plate to facilitate the natural outflow of leachate under the action of gravity. Install an aeration pipe on the side of the column and connect it to the aeration pump. There is a leachate discharge port (10 mm) at the bottom of LSRs, which uses a valve to control the outflow of leachate. The generated leachate flows into the leachate regulating tank from the collection chamber, undergoes pH adjustment, and then is recirculated to LSRs through a peristaltic pump. The effective filling height for the landfill simulation experiment is 600 cm, and a layer of 80 mesh nylon mesh and a 5 cm gravel layer are laid on the upper layer of the perforated plate to prevent leachate from blocking the pipeline.

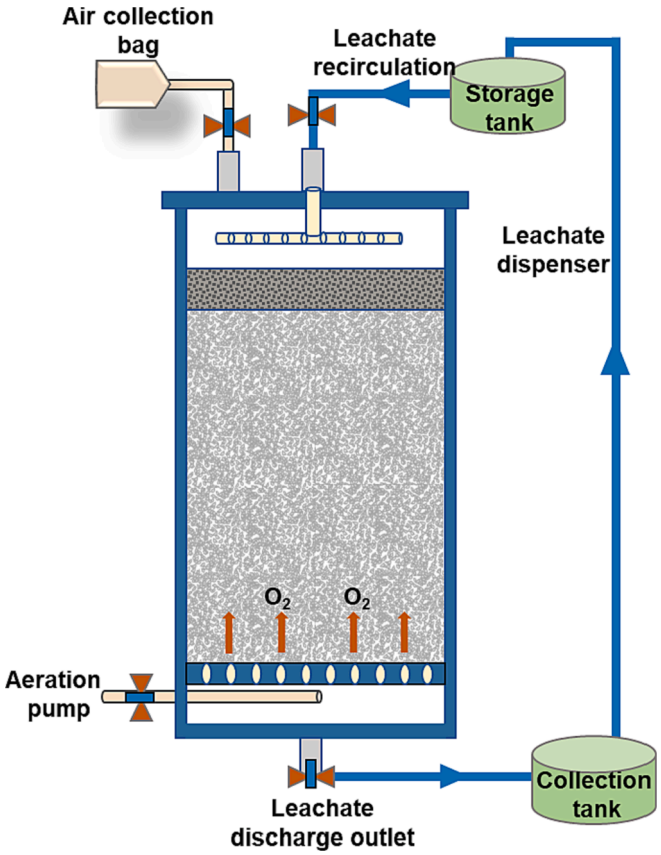


Fig. 1. Landfill simulation reactor (LSR) experimental devices.

The LSR-1 operated without leachate recirculation and aeration, representing conventional landfills. The LSR-2 was subjected to micro-aeration for 15 min at an aeration rate of 100 L/h. LSR-3 was responsible for effluent recirculation. The frequency of recirculation was three days per time, and the volume of recirculated leachate was 2 L. The LSR-4 was treated with micro-aeration and recirculation of leachate. The aeration rate is 100 L/h, and the aeration time is 15 min; the frequency of leachate recirculation is 3 d/times, the recirculation volume was 2 L, and the recirculation time was 20 min. According to Table 2, each landfill column is operated according to its own process parameters.

2.3. Analytical methods

Water content was determined using oven drying at 105 °C for 12 h to determine weight difference, and organic matter was scorched using a muffle furnace KSL-1100X at 600 °C, before and after weight difference. Conventional indicators such as COD, TP, and NH_4^+–N were quantified according to the APHA standard method (Leckie et al., 1979). VFA was determined according to gas chromatography (GC-2014 (Shimadzu)) and pH was analyzed by a pH meter (PHSJ-4F (Leica, China)). UV_{254} was defined as the UV absorbance of the water sample at 254 nm and was determined using a Hach DR-5000 UV–visible spectrophotometer (Hach, USA) to determine the UV–visible spectrum of the leachate, scanning at 200–700 nm. The methane (CH_4) and carbon dioxide (CO_2) contents and the biogas produced in the subsequent anaerobic phase were measured by a gas chromatography (GC-2014 (Shimadzu)) equipped with a thermal conductivity detector.

2.4. DNA extraction and 16S ribosomal RNA gene quantitative PCR (qPCR)

Samples were collected from the middle and bottom of the landfill pile on days 5, 105, and 350 and mixed in equal amounts for subsequent DNA extraction. DNA was extracted using the EZNA® Soil DNA Kit (Omega Bio-Tek, Norcross, GA, US). DNA concentration and purity were checked using a NanoDrop 2000 UV-V is spectrophotometer, and DNA extraction quality was checked using 1 % agarose gel electrophoresis. For PCR amplification, primer pairs 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′) were used to amplify the bacterial 16S gene. Primer pairs 524F10extF (TGYCAGCCGCCGCGGTAA) and Arch958RmodR (YCCGGCGTTGAVTCCAATT) were used to amplify against archaea. Through the use of the MiSeq PE300 platform, the final high-throughput sequencing was performed by Majorbio Bio-Pharm Technology Co. (Shanghai, China). Analyzing the obtained data was performed using the QIIME II procedure. A detailed description of the procedure can be found in a previous study (Lv et al., 2021). Raw sequenced sequences were quality controlled using Trimmomatic software and spliced using FLASH software. Sequences were ASV clustered based on 97 % similarity using UPARSE software. Chimeras were eliminated using UCHIME software. Species classification was annotated for each sequence using RDP Classifier with 70 % confidence threshold. The alpha diversity of each

Table 2
Operating parameters of the experimental device.

Number	Waste pile	Leachate	Operation mode	Notes
1	No aeration	No recirculation	Sealed anaerobic condition	Control group
2	Aeration	No recirculation	Aeration frequency 3 d/ times	
3	No aeration	Recirculation	Recirculation frequency 3 d/times,	
4	Aeration	Recirculation	Recirculation frequency 3 d/timesaeration frequency 3 d/times	

sample was analyzed using MOTHUR software as Coverage index.

2.5. Data analysis

The structure of the microbial community and environmental variables in landfill were analyzed using a Monte Carlo test based on 999 random permutations. The SPSS (Statistical Package for the Social Sciences) Statistics 22.0 software (IBM, USA) was used for statistical analysis. One-way ANOVA and Bonferroni post-tests were used to compare the data difference, and the evaluation criterion of significant difference was $p < 0.05$.

3. Results and discussion

3.1. Stabilization efficiency of landfill waste

3.1.1. Solid phase degradation process

Organisms require a continuous absorption water from the surrounding environment to maintain their growth and metabolic activities (Kim et al., 2017). The high moisture content of waste helps microbial metabolic processes and increases the rate of waste degradation (Huang et al., 2022). The waste with different landfill structures maintained a high moisture content (greater than 45 %) (see supplementary materials). The initial water contents of LSR-1, LSR-2, LSR-3 and LSR-4 at the beginning of the experiment were 55.56 %, 55.77 %, 55.67 % and 56.87 %, respectively. In addition, the moisture content was critical to microbial activity, and LSR-1 lacking sufficient leachate recirculation and oxygen entry, resulted in low water content in the waste, which could have a detrimental effect on the biodegradation of organic matter. Based on the variation of LSR-3, increasing recirculation volume within a specific range had shown to be an effective strategy for reducing evaporation and accelerating waste degradation without affecting moisture content. Until terminal phase of the landfill experiment, the water contents of LSR-1, LSR-2, LSR-3 and LSR-4 were 39.26 %, 41.2 %, 44.01 % and 45.07 %, respectively. The higher water content was conducive to the metabolism of microorganisms, the migration and transformation of pollutants in the landfill.

Organic matter content is related to the biodegradability and stability of landfill waste (Peng et al., 2022). Insoluble organic matter such as fats, proteins and polysaccharides in the solid phase organic matter are hydrolyzed to long-chain fatty acids, amino acids and soluble sugars, respectively, by extracellular enzymes (Ishigaki et al., 2004). Usually, the hydrolysis rate of sugars is faster than that of proteins and fats, and thus the hydrolysis of proteins and fats plays a controlling role in landfill stabilization, becoming the rate-limiting stage of the whole bioreactor landfill stabilization process (Prantl et al., 2006). The organic matter content in the initial stages of all four reactors was around 56 % (see supplementary materials). After the landfill experiment, the organic matter in LSR-1, LSR-2, LSR-3, and LSR-4 fluctuated downward to 45.1 %, 39.16 %, 38.97 %, and 41.38 %, respectively. Throughout the experiment, the waste in LSR-1 showed the least decrease in organic matter content, mainly due to the LSR-1 column being an anaerobic bioreactor, which created a low pH environment at the initial stages of the landfill experiment, inhibiting hydrolysis bacteria, which affected the hydrolysis rate of landfill waste. LSR-3, although an anaerobic unit, was recirculated by leachate, and the leachate recirculation contained suitable pH and certain nutrients, the microorganisms accelerated the degradation of macromolecular organic matter under this suitable environment. From the analysis of the changing trend of organic matter, the degradation effect of landfill waste under anaerobic conditions (LSR-1) was the worst. In contrast, LSR-4 with certain leachate recirculation and micro-aeration techniques enhanced organic matter degradation within the landfill. The results showed that the combined air–water conditioning technology of micro-aeration and leachate recirculation had a positive effect on accelerating the degradation of organic matter in landfill waste.

3.1.2. Degradation of pollutants in leachate

The concentration of COD in the leachate generated from the initial landfill fluctuated greatly from 8000 to 13000 $mg{\cdot}L^{-1}$, as shown in Fig. 2 (a). After 50 days of the landfill, the COD concentration in the leachate of each LSR showed a rapid increase, with values reaching 100500 $mg{\cdot}L^{-1}$, 921600 $mg{\cdot}L^{-1}$, 90200 $mg{\cdot}L^{-1}$, and 876500 $mg{\cdot}L^{-1}$ for LSR-1, LSR-2, LSR-3, and LSR-4 landfill columns, respectively. This was mainly due to the decomposition and acidification of the waste in the presence of oxygen to produce monosaccharides, amino acids, and glycerol. Thereafter, the concentration of COD continued to rise, the reaction entered the acidification phase, the production of VFAs increased, and the pH decreased. At the 350 d of the experiment, the COD values of the four LSRs were 42587 $mg{\cdot}L^{-1}$, 29412 $mg{\cdot}L^{-1}$, 35754 $mg{\cdot}L^{-1}$, and 27754 $mg{\cdot}L^{-1}$, respectively. The concentration of COD in LSR-4 was significantly lower than that of other LSRs due to the conditioning effects of micro-aeration and leachate recirculation, which provided a suitable environment for microbial growth and promoted the production of large amounts of extracellular hydrolytic enzymes by hydrolytic acidifying bacteria, thus accelerating the leaching of COD from the landfill into the leachate. The trend of COD of LSR-2 and LSR-3 at the 350 d of the experiment showed that the COD concentration of leachate recirculation was greater than that of the micro-aeration group. Leachate recirculation helped to increase the leaching and transfer of organic matter in landfills, providing a suitable growth environment for microbial degradation of organic matter. And after accounting for the carbon balance of the solid–liquid-gas phases of the landfill pile, the residual carbon contents of the four samples were 59.3 % (LSR-1), 46.2 % (LSR-2), 347 % (LSR-3), and 24 % (LSR-4), and it was found that the solid-phase carbon transfer to the gas-phase carbon of LSR-4 was 60 %, which is higher than that in LSR-1, which was the complete control, and the transfer of solid-phase carbon was 40 % (see Supplementary Material). Micro-aeration and leachate recirculation of LSR-4 has been shown to rapidly degrade organic matter during the hydrolysis acidification stage, promoting the rapid consumption of products by microorganisms in the hydrolysis acidification stage, thereby converting organic matter into gaseous form.

Phosphorus in wastewater exists mainly in the form of primary phosphate, and a small amount of organic phosphate exists mainly in the form of protein. The elemental phosphorus in landfill waste is relatively stable, and only a small amount of phosphorus-containing material is decomposed into soluble metaphosphate (HPO_4^{2-}) by relevant microorganisms. As shown in Fig. 2(b), the TP concentrations in the four new landfill reactors were 262 $mg{\cdot}L^{-1}$, 241 $mg{\cdot}L^{-1}$, 240 $mg{\cdot}L^{-1}$, and 261 $mg{\cdot}L^{-1}$, respectively. Due to the relatively low phosphorus demand of anaerobic microorganisms, the control group had a constant TP concentration. The TP in LSR-1 was reduced by 20.75 %, and a higher reduction was not achieved. Until the 350 d of the landfill experiment, the total phosphorus concentrations in the four LSRs were 218 $mg{\cdot}L^{-1}$, 197 $mg{\cdot}L^{-1}$, 182 $mg{\cdot}L^{-1}$, and 152 $mg{\cdot}L^{-1}$. The micro-aeration process conditions had lower TP concentrations than the leachate recirculation conditions, and the waste piles were in a partly aerobic and anaerobic environment with fast microbial degradation of organic matter. The TP removal rate of LSR-4 after intermittent micro-aeration and leachate recirculation (48.3 %) was higher than the other experimental groups.

UV_{254} is a parameter used to determine the degree of aromatization of organic matter present in leachate (Renou et al., 2008). In the initial stage, the UV_{254} of leachate decreased with leachate recirculation and leaching of organic matter during landfill waste degradation (Fig. 2(c)). The UV_{254} of leachate in LSR-1, LSR-2, LSR-3 and LSR-4 and the reactor reached a maximum on 50 d: 56.6 $L{\cdot}(mg{\cdot}m)^{-1}$, 53.83 $L{\cdot}(mg{\cdot}m)^{-1}$, 48.85 $L{\cdot}(mg{\cdot}m)^{-1}$ and 45.60 $L{\cdot}(mg{\cdot}m)^{-1}$. As organic matter degraded in leachate, the UV_{254} values gradually decreased, indicating the stabilization of landfill waste. The UV_{254} values in LSR-1, LSR-2, LSR-3 and LSR-4 were 17.6 $L{\cdot}(mg{\cdot}m)^{-1}$, 12.13 $L{\cdot}(mg{\cdot}m)^{-1}$, 9.85 $L{\cdot}(mg{\cdot}m)^{-1}$ and 9.85 $L{\cdot}(mg{\cdot}m)^{-1}$, respectively.

The changes in the concentration of NH_4^+–N in the leachate were

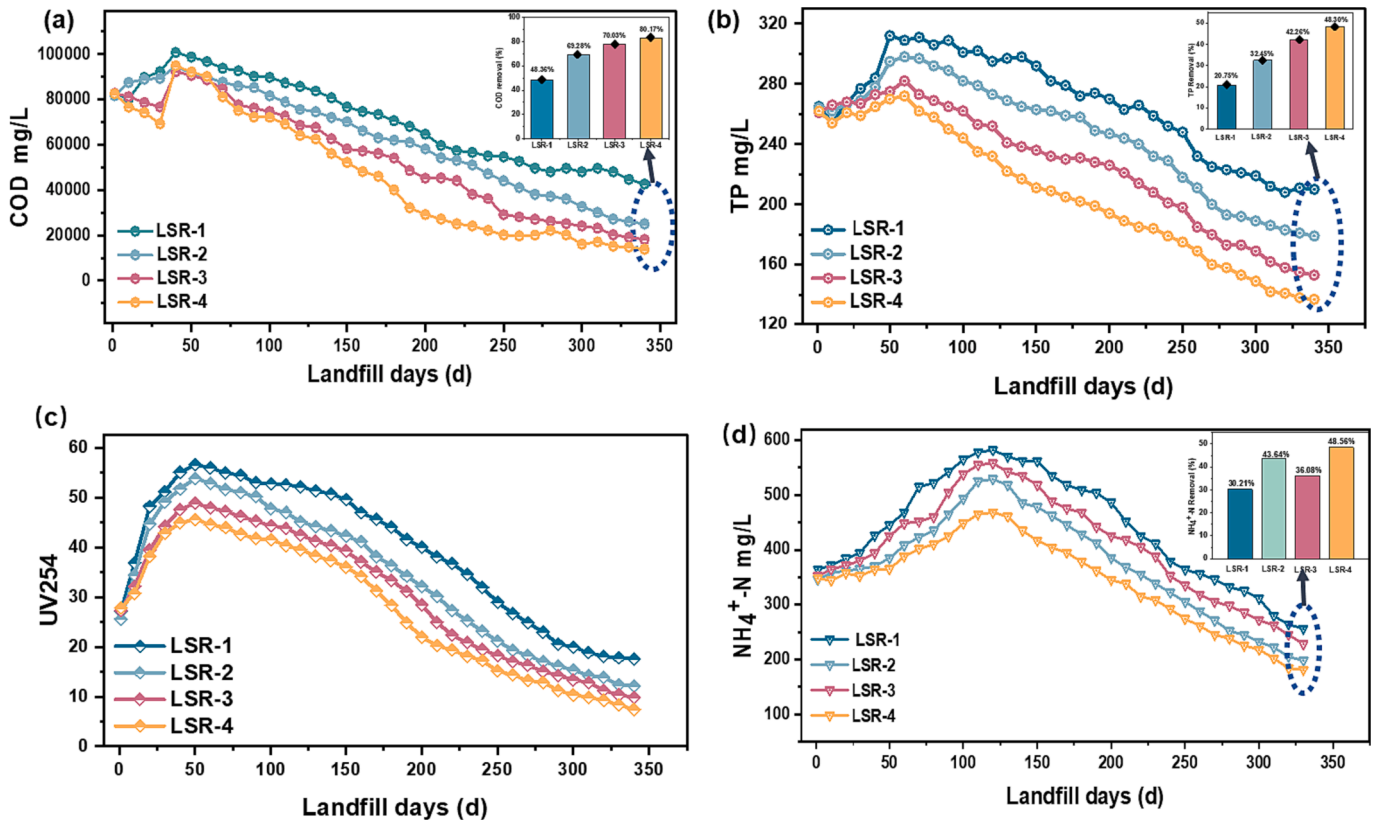


Fig. 2. (a) Change of COD concentration in leachate; (b) change of TP concentration in leachate; (c) Change of UV_{254} concentration in leachate; (d) change of NH_4^+–N.

displayed in Fig. 2(d), where the concentration of NH_4^+–N showed an increasing trend followed by a decreasing trend before finally remaining relatively stable. Due to the presence of a small amount of oxygen at the beginning of the landfill, a substantial amount of NH_4^+–N was generated by the degradation of proteins and amino acids, resulting in an increase in the concentration of NH_4^+–N (Wu et al., 2014). By the 110 d of landfill experiment, the concentrations of NH_4^+–N reached maximum values in LSR-1, LSR-2, LSR-3 and LSR-4: 578 mg·L^{-1}, 525 mg·L^{-1}, 555 mg·L^{-1} and 465 mg·L^{-1}. After 110 d of the landfill, NH_4^+–N was produced as NO_2^-–N by nitrite bacteria, and then NO_2^-–N was converted to NO_3^-–N by nitrate bacteria, and the effluent NH_4^+–N concentration also decreased compared with the influent water. Under anaerobic conditions, NO_3^-–N was removed by the action of denitrifying bacteria, which were converted to nitrogen gas. The NH_4^+–N concentration in LSR-1, LSR-2, LSR-3 and LSR-4 were: 254 mg·L^{-1}, 195 mg·L^{-1}, 225 mg·L^{-1} and 179 mg·L^{-1} at the end of landfill. The removal of NH_4^+–N mainly occurred via the simultaneous nitrification–denitrification pathway. Under anaerobic conditions (LSR-1), nitrogen was selectively converted to ammonia, which accumulated in the leachate due to the lack of available metabolic pathways. LSR-4 removed NH_4^+–N through denitrification due to the leachate circulation effect, which could accelerate the rate of ammonification inside the landfill reactor. It was observed that the concentration of NH_4^+–N in the reactor after micro-aeration (LSR-2) was lower than that after leachate recirculation (LSR-3). This observation indicated that micro-aeration alone was performed, and nitrification occurred inside the landfill pile, which had a positive effect on denitrification. After accounting for the nitrogen balance in the leachate (see attached material), it was found that the NH_4^+–N /TN (total nitrogen) in LSR-4 (39.1 %) was higher than the NH_4^+–N/TN in LSR-1 (32.9 %), and that the NO_3^-–N/TN in LSR-4 (12.8 %) was higher than the NO_3^--N/TN in LSR-1 (5.6 %). The results also proved that the supply of trace oxygen after the positive promotion of nitrogen conversion.

3.1.3. Gas composition

In terms of gas composition, landfill gas production primarily results from the degradation of organic matter in landfill waste. Gas production results after 350 d of landfill experiments showed that micro-aeration and leachate recharge (LSR-4) had the highest cumulative gas production (see supplementary material). Fig. 3 showed the yield and gas composition of CO_2 and CH_4 gases in the landfill gas produced by each reactor. The CO_2 emissions from each reactor exhibited an increasing and then decreasing trend. Around 150 d of the landfill, the CO_2 production of each group reached the highest peak with the concentration of CO_2 at 76.6 %, 74.95 %, 68.91 % and 73.73 %, respectively. The methane emission of each reactor showed a trend of gradually increasing and then slowly decreasing. The maximum methane production was also concentrated around 150 d, but its peak concentration occurred at the 350 d of the landfill, with 43.91 %, 54.22 %, 52.22 % and 59.22 % for LSR-1, LSR-2, LSR-3 and LSR-4, respectively. The micro-aeration and leachate recirculation group (LSR-4) could enter the methanogenic stage faster than the other experimental groups, and the methanogenic lag time was shorter. The results indicated that micro-aeration can promote the hydrolytic acidification of aerobic microorganisms, resulting in the production of large amounts of CO_2. With the gradual depletion of O_2 in LSR-4, the landfill pile entered a facultative anaerobic environment and the facultative microorganisms played the main dominant role in decomposing small-molecule acids, which also produced large amounts of CO_2, but their production was not as large as that of the hydrolytic acidification stage. When LSR-4 entered the methanogenic stage, the CH_4 concentration was higher than the CO_2 concentration, and the CH_4 concentration increased gradually and gently with the landfill time, indicating that the reactor entered the stable methanogenic stage. The methanogenic phase appeared earlier in LSR-2 and LSR-3 compared to controls. And the methanogenic stage was earlier for the landfill reactor that underwent leachate recirculation

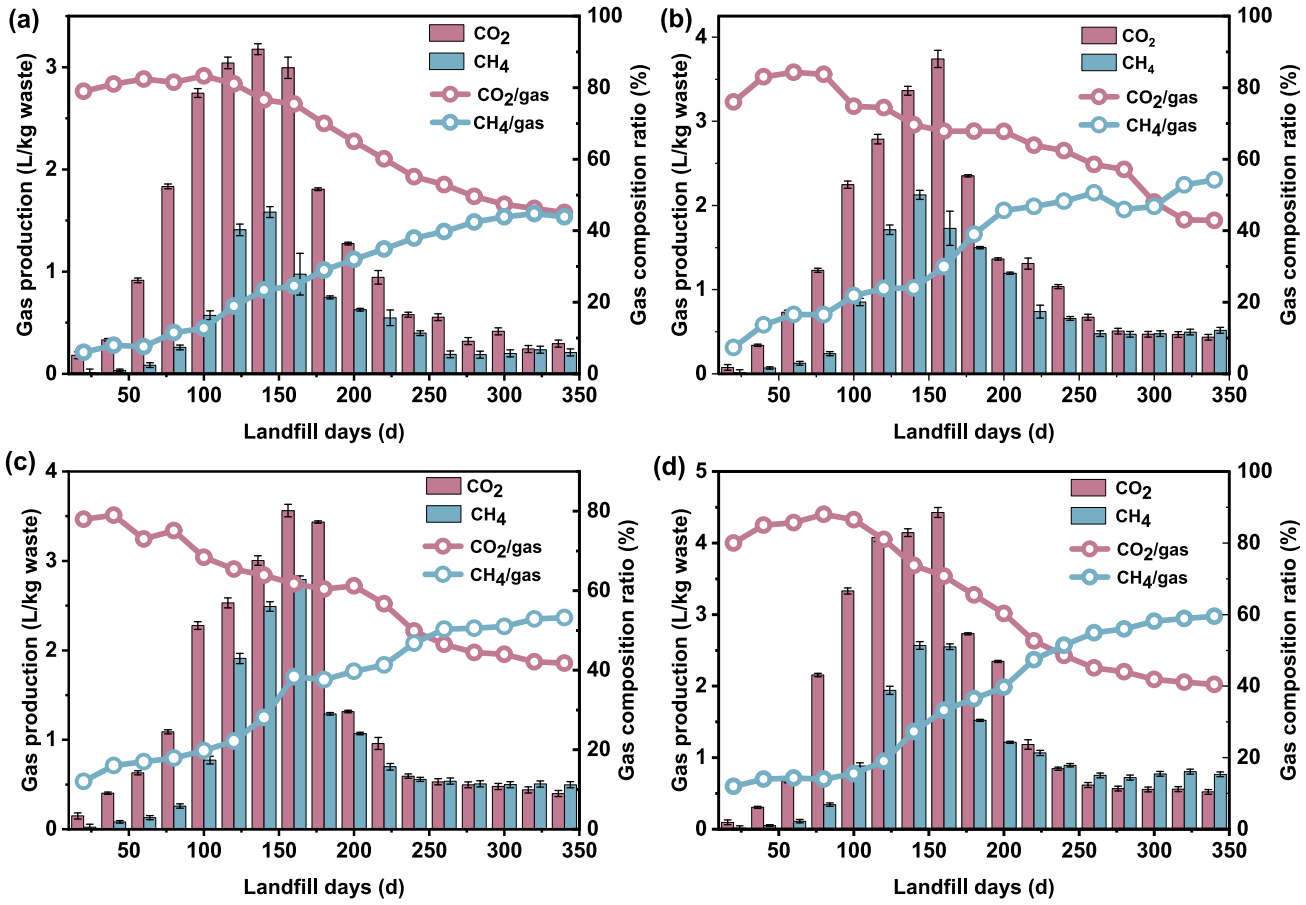


Fig. 3. Gas production diagram and gas composition of LSR-1 (a); LSR-2 (b); LSR-3 (c); LSR-4 (d).

(LSR-3) than that for the landfill reactor that underwent micro-aeration (LSR-2). Because the recirculation of leachate provided suitable moisture for microorganisms, it promoted the circulation of nutrients and related hydrolytic enzymes in the landfill site, which increased methane production. Micro-aeration provided a partial anaerobic environment for the landfill, it was beneficial for the growth of facultative microorganisms. However, the integrated process of micro-aeration and leachate recirculation promoted an increase in methane and carbon dioxide production, which helped to accelerate the stabilization process of landfill waste.

3.2. *Hydrolysis acidification of integrated process*

The integrated process group of micro-aeration and leachate recirculation had a strong advantage in hydrolysis acidification compared to the experimental control group. The organic matter content (41.5 %) in the integrated process group on the 150 d was lower than that in the control group (51.3 %), indicating that the biodegradation efficiency of landfill waste in the integrated process group was higher (see supplementary materials). The catabolism of extracellular hydrolytic enzymes within the cell accelerated the conversion of large to small organic molecules (Li et al., 2022a, Li et al., 2022b). During the period of 0–150 d, the organic matter of landfill waste in the integrated process decreased by 18 %, while that in traditional landfill decreased by 12 %. This also indicated that micro-aeration and leachate recirculation had strong advantages in accelerating the efficiency of the hydrolysis stage.

The accumulation of VFAs during the process of waste stabilization is a prominent concern for current research experts. VFAs not only serve as products of the hydrolysis acidification stage but also act as substrates for the methanogenic stage (Lay et al., 2014). If VFAs accumulate in substantial quantities, it can not only lower the pH value but also severely disrupt the equilibrium of the hydrolysis-acidification stage, impeding the methanogenesis process (Matsuto et al., 2015). Moreover, the long-term accumulation of acidic substances seriously affects the survival of methanogenic bacteria, which makes the methanogenic stage seriously delayed and inhibits the stabilization of landfill (Hettiaratchi et al., 2014). Consequently, this study investigated and analyzed the relationship between pH and VFAs to examine the mechanism for micro-aeration and leachate recirculation to accelerate the stabilization process of landfill waste.

The monitoring of leachate pH constitutes one of the key parameters for determining the stage of waste degradation (Vigil et al., 2015). As illustrated in Fig. 4(a), the pH values were low (4–6) at the beginning of the landfill due to the acidification and hydrolysis process of landfill waste. According to Fig. 4(b), the accumulation of VFAs peaked at around 150 d. The lower pH of the landfill tended to increase the proportion of free VFAs, while the H^+ dissociated from the free-state VFAs into the cells could lead to cytoplasmic acidification. Fig. 4(c) revealed that acetic acid content reached its maximum value at about 100 d of the landfill, owing to the dominant role played by hydrolytic and acidifying bacteria in the production of a substantial amount of acetic acid from macromolecular organic matter (Chung et al., 2015). In comparison with other experimental groups, LSR-4 showed a rapid increase in acetic acid content, entering the stage of hydrogen and acetic acid production first, thereby demonstrating that micro-aeration and leachate recirculation helped to accelerate acidification within landfill waste. During the period of 150–200 d in the landfill, the pH values of LSR-1, LSR-2, LSR-3 and LSR-4 rapidly reached 7.0, 6.61, 6.81, 6.93 and 7.20, respectively. The total volatile fatty acid (TVFA) content of each LSR in this stage gradually decreased, the hydrolytic acidification stage tended to end,

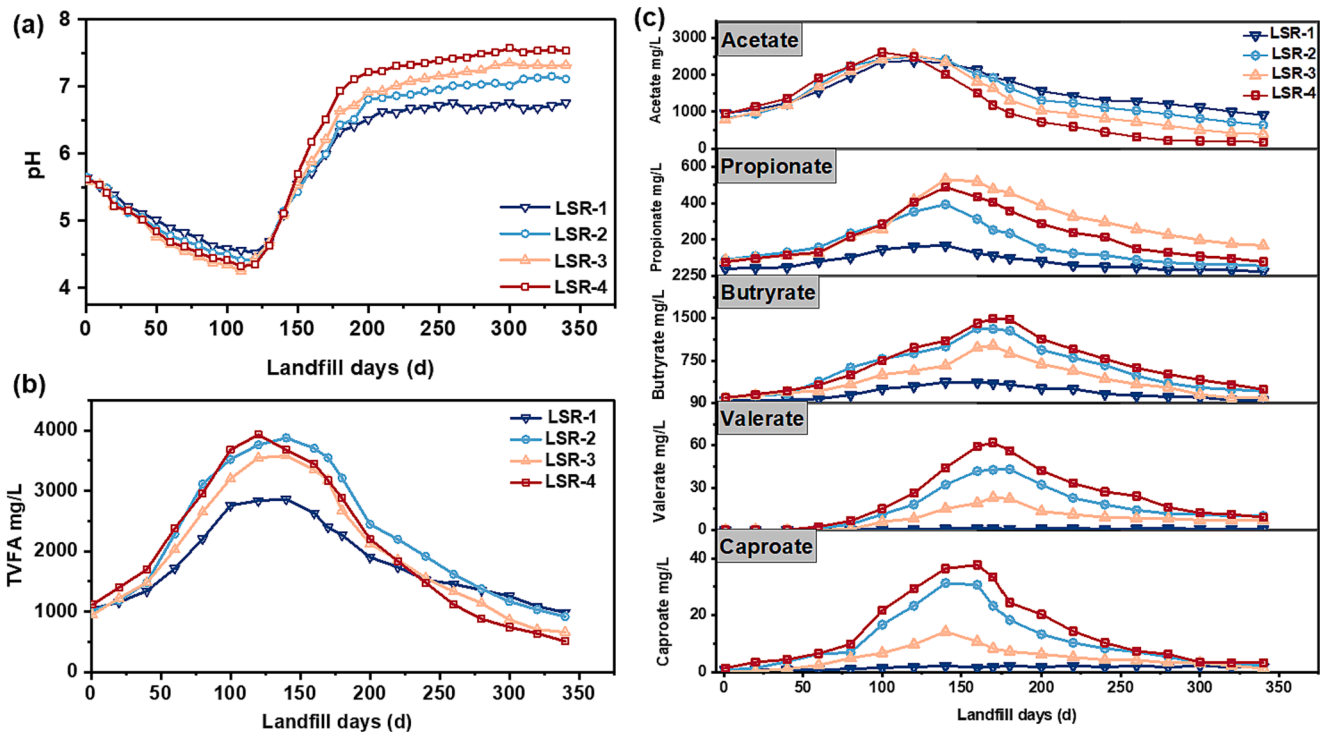


Fig. 4. (a) Change of pH; (b) total volatile fatty acid in the reactor; (c) concentration changes of acetate, propionate, butyrate and valerate.

the content of substrate available to methanogenic bacteria gradually decreased, and the LSR entered the methanogenic stage. As shown by Fig. 4(c), the butyric and capric acid contents of LSR-2 and LSR-4 increased, indicating that micro-aeration may facilitate the rapid conversion of acetic acid into butyric and capric acids by facultative bacteria (e. g. *Clostridium_sensu_stricto_12*). From 200 d to the 350 d of the landfill, the pH values of the LSR-2, LSR-3 and LSR-4 had reached a neutral level of 7.01, 7.15 and 7.27, respectively. The pH value of LSR-1 remained lower throughout the experiment until the 350 d of the experiment when it reached 6.76. Acidity was the main inhibitor of methane production under anaerobic conditions. The low pH of LSR-1 indicated that the long-term dominance of hydrolytic and acid-producing bacteria allowed the accumulation of small molecules of organic acids, which delayed the onset of the methanogenic phase. The pH of LSR-4 after micro-aeration and recirculation were significantly higher than that of other LSRs, which was favorable for facultative bacteria to promote the degradation of VFAs and increase the pH to facilitate the methanogenic phase.

The concentration of VFAs is a crucial intermediate metabolite in the anaerobic landfill (Yu et al., 2022). Fig. 4(b) showed the accumulated VFAs concentration in each LSR during the landfill process, and the VFAs concentration can reflect the degree of hydrolysis and acidification of substrates in the system as well as the efficiency of methanogenesis. The micro-aeration and leachate recirculation conditions (LSR-4) produced a large amount of TVFA in a short period, which provided part of the substrate for the methanogenic process. It was possible that the long-term accumulation of small organic acids delays the onset of methanogenesis as a result of the low pH of LSR-1. And the leachate recirculation provided moisture regulation, which enhanced the action of acetobacter and methanogenic bacteria to fully utilize a large amount of VFAs (Lay et al., 2014). The intermediate products that can be directly used by methanogenic bacteria were formic acid, methanol, acetic acid, H_2 and CO_2 (Wu et al., 2014). The accumulation of other long-chain fatty acids can lead to acidification of the system, which is detrimental to the methanogenesis process. As shown in Fig. 4(c), acetic acid was the main fatty acid in the landfill process and the concentration of acetic acid dropped to the lowest after entering the methanogenic stage of stabilization after 200 d. Additionally, in line with earlier research, the results from Fig. 5(a) demonstrated that micro-aeration and leachate recirculation (LSR-4) increased the relative abundance of hydrolytic and fermentative bacteria (*Proteobacteria*, *Chloroflexi*, and *Actinobacteria*) (Bandala et al., 2021). *Lactobacillus*, the most abundant lactic acid bacteria in the landfill reactor in this study, could utilize a variety of carbohydrates such as cellulose and lignin from landfill waste such as xylan (Grossule and Lavagnolo, 2020), mainly producing acetic acid, which corresponded to the rapid increase in acetic acid content for a short time in Fig. 4(c). The results also indicated the positive role of micro-aeration and leachate recirculation in promoting the hydrolytic acidification phase. As the abundance of *Pseudomonas* in LSR-4 increased, it was able to generate large amounts of hydrogen gas from acetic acid, thus enabling rapid degradation of acetic acid. Studies have revealed that methanogenic bacterial growth was significantly inhibited by the concentration of propionic acid (Wang et al., 2022). In this study, propionic acid slowly decreased to lower concentrations after 150 days and did not inhibit methanogenic bacteria. As for valeric and capric acid, their levels were so small that they had no significant effect on the landfill system and degraded almost completely during stable operation.

3.3. Microbiological mechanism of landfill stabilization

Samples with different landfill times were selected and the microbial communities in the four LSRs were analyzed in different dimensions. The correlations between microbial community structure, integrated process operating parameters and environmental factors were explored, and the mechanism of microbial degradation of landfill waste was deeply explored.

3.3.1. Enrichment of facultative hydrolytic bacteria

As depicted in Fig. 5(a), *Firmicutes* constituted the most prevalent bacterial phylum with a relative abundance ranging from 56 % to 76 % throughout all landfill phases. *Proteobacteria* (10 %–31 %) and *Actinobacteriota* (3 %–6 %), *Bacteroidota* (2 %–8 %), *Patescibacteria* (1 %–6 %),

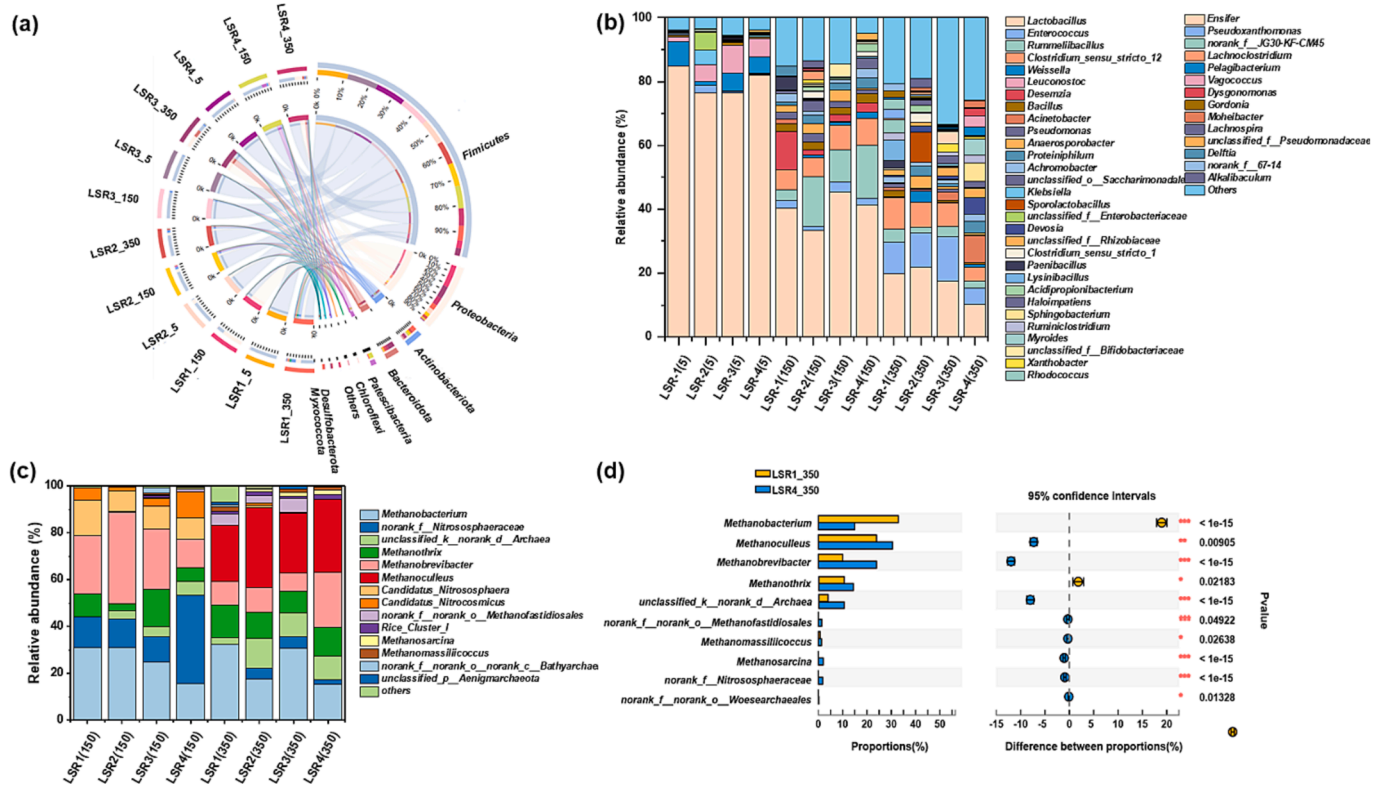


Fig. 5. Analysis of bacterial and archaeal community structure: (a) bacterial communities at the phylum level; (b) bacterial communities at the genus level; (c) community structure at the genus level of archaea; (d) differences at the genus level of archaea in LSR-1 and LSR-4 at 350 d.

and *Chloroflexi* (0.2 %–2 %) were the subsequent phyla observed. While *Proteobacteria* convert soluble sugars into monosaccharides and short-chain fatty acids, *Firmicutes* and *Bacteroides* mostly break down hydrolyzed cellulose and starch polysaccharides in landfill (Wu et al., 2016). *Firmicutes* bacteria proliferated most rapidly in the waste at the outset of the landfill, which may have provided sufficient carbohydrate and monosaccharide substrates for the growth and proliferation of *Proteobacteria*. According to Fig. 5(a), the abundance of the *Firmicutes* was high at the beginning of the landfill reactor, accounting for about 12 %. However, the increase in landfill time led to a gradual decrease in the abundance of the *Firmicutes*. Until the 350 d of the landfill, the abundance of *Firmicutes* in the LSR-1, LSR-2, LSR-3 and LSR-4 were 34.3 %, 45.4 %, 45.3 % and 23.6 %, respectively. The decrease in abundance of the *Firmicutes* also represents a greater enrichment of other species, such as *Proteobacteria* and *Bacteroidetes*. It is believed that *Proteobacteria* and *Bacteroidetes* are crucial for the breakdown of organic materials and the cycling of carbon (Nghiem et al., 2017). Bacteroidota are recognised for their versatile heterotrophic metabolism of carbohydrates such as lignin, cellulose, and some difficult-to-degrade organic matter (i.e. humic substances). Conversely, the relative abundance of *Bacillus* in LSR-4 was initially low; however, with intermittent aeration and leachate recirculation in the LSR for environmental regulation, *Bacillus* became more enriched and maintained a stable proportion of 25.5 %. The relative abundance of other phyla, including *Chloroflexi*, *Acidobacteria*, and *Desulfobacterota*, was extremely low, <1 %. *Desulfobacteria* can utilise numerous organic compounds, including alcohols and volatile fatty acids. *Desulfobacteria* showed some enrichment in the late landfill stage of LSR-4, indicating that leachate recirculation and micro-aeration facilitated the conversion of VFAs and the stabilisation of landfill waste. However, in other reactors, the abundance of *Desulfobacterota* remained at low levels (<1 %). Furthermore, the *Desulfobacterota* phylum was also tested for its DIET ability, allowing it to perform DIET with methanogenic bacteria concurrently, enhancing the electron transfer efficiency and the methanogenic capacity of methanogenic bacteria in landfills.

To further comprehend the impact on the population after micro-aeration and leachate recirculation and the trend of the population, Fig. 5(b) illustrated the distribution characteristics of microorganisms at the genus level in the four LSRs. The species abundance in all reactors increased with landfill time, and especially the community richness in LSR-4, where leachate recirculation and micro-aeration were performed, had a more apparent increase compared to the other experimental groups. The abundant bacterial community is believed to enhance the efficiency of the system in treating landfills through inter-population coordination, thus promoting the stability of the landfill system operation.

A total of 44 genera with abundance ratios < 0.02 were detected in this bacterial examination, among which the dominant genera as the top 10 in overall abundance were *Lactobacillus*, *Enterococcus*, *Rummeliibacillus*, *Clostridium_sensu_stricto_12*, *Weissell*, *Leuconostoc*, *Desemzia*, *Bacillus*, *Acinetobacter*, *Pseudomonas* and other dominant genera. At the beginning of the landfill (5 d), the absolute dominant genera of microbial community composition were *Lactobacillus* (76 %–84 %), *Weissella* (5 %–7 %), and *Leuconostoc* (4 %–8 %). The absolute dominant genus *Lactobacillus*, which belongs to Gram-positive bacteria, has a strong ability to metabolize carbohydrates to produce acid (Gou et al., 2014). *Lactobacillus* can synthesize glucan and heteropolysaccharides and can make sugar fermentation to produce lactic acid or other acid substances (Zhu et al., 2011). Moreover, the pH during this stage was low, primarily due to acetic acid production and slightly low biodiversity. There was no significant microbial community difference among the four simulated experimental groups at the initial stage of the landfill. At the middle stage of the landfill (150 d), the biodiversity was richer and the community structure with *Clostridium_sensu_stricto_12* (6.111 %–8.376 %) and *Rummeliibacillus* (3.318 %–15.587 %) as the dominant populations emerged. *Clostridium_sensu_stricto_12* was shown to metabolize ethanol and lactic acid to produce hexanoic acid and butanoic acid. And the abundance of *Clostridium_sensu_stricto_12* was lower in LSR-4 than in the

other three groups, indicating that the landfill environment of LSR-4 has made the accumulation of butyric and caproic acid concentrations much weaker, which was favorable for the methanogenic phase. *Rummeliibacillus* under micro-aeration and leachate recirculation conditions appeared to be heavily enriched compared to the other experimental groups, and the bacterium secreted a variety of cellulases and hemicellulases. The anaerobic bacteria *Proteiniphilum* and facultative anaerobic bacteria *Achromobacter* were also observed to be significantly enriched in the process of waste degradation, and *Proteiniphilum* was also proved to have the function of hydrolysis and acidification. The presence of *Achromobacter* proved that the environment in the landfill was more suitable for the growth and survival of parthenogenic anaerobic bacteria (Gu et al., 2019). At the end of the landfill (350 d), the species richness of *Enterococcus* (5.112 %–14.043 %) and *Clostridium_sensu_stricto_12* (4.209 %–9.873 %) was higher. *Enterococcus* are Gram-positive bacteria (G^+) with a strong hydrogen production capacity, which can reduce the environmental requirements of the fermentation hydrogen production system and improve the economic efficiency of the whole system. The landfill entered the late methanogenic stage proceeding, and this stage had the most abundant microbial community structure, which also indicated that the community structure was more stable and balanced, contributing to the stabilization process of the entire landfill. *Clostridium_sensu_stricto_12* is a strictly anaerobic microorganism, whereas most *Bacillus* are parthenogenic or aerobic microorganisms. As the landfill transitioned to a strictly anaerobic state, the proportion of *Clostridium_sensu_stricto_12* in LSR-1 gradually increased, while the proportion of *Bacillus* decreased. The species richness of *Bacillus* was higher in LSR-2, and since *Bacillus* was facultative or aerobic bacteria, its enrichment also proved that the environment within the landfill was favorable for facultative bacteria to grow and metabolize the short-chain fatty acids accumulated inside the pile, facilitating the beginning of the methanogenic phase. *Pseudomonas* can metabolize a variety of organic compounds such as amino acids, sugars, aromatic hydrocarbons and organic acids (Lv et al., 2021). However, organic compounds such as amino acids and sugars were progressively decomposed and consumed, and the carbon and energy sources required for the growth of *Pseudomonas* decreased, resulting in a decreasing trend in the proportion of *Pseudomonas*.

Compared with LSR-1, the number of *Pseudomonas* in the LSR-4 group of this experiment was the lowest, indicating that it was more stabilized than the experimental group and the landfill waste already had few metabolizable macromolecular organic substances. The detection of *norank_JG30-KF-CM45* as a typical denitrifying bacterium during the experiment confirms the presence of denitrification in the landfill. The process facilitated the microbial reduction of nitrate and nitrite to gaseous nitrogen compounds and nitrogen gas, thereby greatly promoting the escape of reactive nitrogen from the landfill pile in the form of nitrogen gas.

Several genera with low abundance were classified as other genera, and the total percentage of other genera is approximately 6–16 %. The less abundant bacterial communities also played an important role in waste degradation and are involved in complex interactions between microbial communities of biodegradation. Bacterial communities with different characteristics played different ecological roles. High-abundance communities mainly contribute to nutrient cycling, carbon conversion and biomass metabolism, while low-abundance bacterial communities play key functional roles and may be more active than high-abundance bacteria (Abuabdou et al., 2020).

3.3.2. Positive effects of archaeal community structure

Among the 14 archaeal genera detected by the universal primers, *Methanobacterium*, *norank_f_Nitrososphaeraceae*, *unclassified_k_norank_d_Archaea*, *Methanoculleus*, *Methanothrix*, *Methanobrevibacter*, and *Methanoculleus* were the six most numerous genera among all samples (Fig. 5(c)). The archaeal primer pairs identified 15 archaeal genera, of which *Methanobacterium*, *Methanoculleus*, *Methanothrix*, *Methanobrevibacter*, *Methanoculleus* all belonged to the phylum *Euryarchaeota* and *Thaumarchaeota*. accounted for 87.2 %–99.9 % of the total archaeal community (Fig. 5(c)). The genus *Methanobacterium* was the most abundant among the 17 genera in the samples of the landfill reactor up to 150, accounting for 15.5 %–31.1 % of the archaeal community, while the genus *Methanoculleus* accounted for the highest percentage (24.1 %–31.2 %) of the samples at the landfill reactor up to 300 d. *Methanoculleus* and *Methanobacterium* are known to be hydrogenotrophic methanogenic bacteria, while *Methanothrix* is an acetic autotrophic methanogenic bacterium (Hamid et al., 2020).

The high abundance of *Methanobacterium* in LSR-1 during the initial phase stage of the landfill was due to the completely anaerobic environment of LSR-1 during the initial to the mid-term process of the landfill, which was suitable for the growth of *Methanobacterium*. It was also observed that LSR-4 had a higher proportion of *norank_f_Nitrososphaeraceae* belonging to the *Thaumarchaeota* phylum (37.9 %), which was due to the formation of a parthenogenic environment inside the landfill due to micro-aeration, and leachate recirculation provided suitable moisture inside the landfill for the aerobic ammonia oxidation process to occur. In addition, the percentage of *Methanobrevibacter* was relatively high in each reactor at 150 d of landfill. *Methanobrevibacter*, as a genus of methanogenic bacteria that can only use CO_2 as a substrate, indicated that CO_2 gas was the main component at 150 d. Although a small number of *Methanothrix* emerged as Acetic acid methanogenic bacteria at this time, the main methanogenic bacteria in the landfill stage were still dominated by hydrogen and CO_2 as the main substrates.

At the conclusion of the landfill, *Methanoculleus*, the prevailing species, represented a greater proportion in four LSRs. The community composition also indicated that *Methanobrevibacter* accounted for a higher proportion of methanogenic bacteria at this stage. This suggests that leachate recirculation and micro-aeration significantly promoted CO_2 production, leading to a large amount of stable gas produced from the rapid decomposition of organic matter (Luo and Wong, 2019). The proportion of *norank_f_Nitrososphaeraceae* with aerobic ammonia oxidation decreased, also because the microbial community in the LSRs at the late stage of landfill was more stable and less oxygen was available to enter the late stage of methanogenesis.

The difference analysis of methanogenic bacteria between LSR-1 and LSR-4 on 350 d was shown in Fig. 5(d). It was observed that all methanogenic bacteria in LSR-4 except *Methanobacterium* were significantly higher than LSR-1, while the relative abundance of acetate trophic methanogenic bacteria was lower. The results showed that leachate recirculation and micro-aeration measurements enriched hydrophilic methanogens and acetate methanogens in the LSR and accelerated the degradation and consumption of large quantities of VFAs, thus contributing to landfill stabilization.

3.3.3. Correlation mechanism analysis

Micro-aeration and leachate recirculation can enhance the community structure of microorganisms in landfill piles, and the degradation ability of contaminants in the landfill stabilization process is inevitably related to the community structure of microorganisms. Micro-aeration and leachate recirculation to promote the degradation ability of pollutants in the landfill reactor can be effectively found. As shown in Fig. 6 (a), the relationship between the degradation ability of contaminants and the community structure of microorganisms was indeed significant. Furthermore, environmental factors such as COD, UV_{254}, VFAs, NH_4^+- N, and acetic acid were positively correlated with *norank_f_Nitrososphaeraceae*, *Methanobrevibacter*, and *Firmicutes*. COD, UV_{254}, VFAs, NH_4^+–N and acetic acid were negatively correlated with *Methanoculleus*, *Actinobacteriota* and *Methanothrix*. The results also showed that as the landfill stabilization process proceeded and the contaminant concentration gradually decreased, the emergence of hydrogenotrophic methanogenic bacteria and CO_2 based substrate methanogenic bacteria as the dominant populations promote the methanogenic process and the stabilization of the landfill reactor. The enrichment of hydrogenotrophic

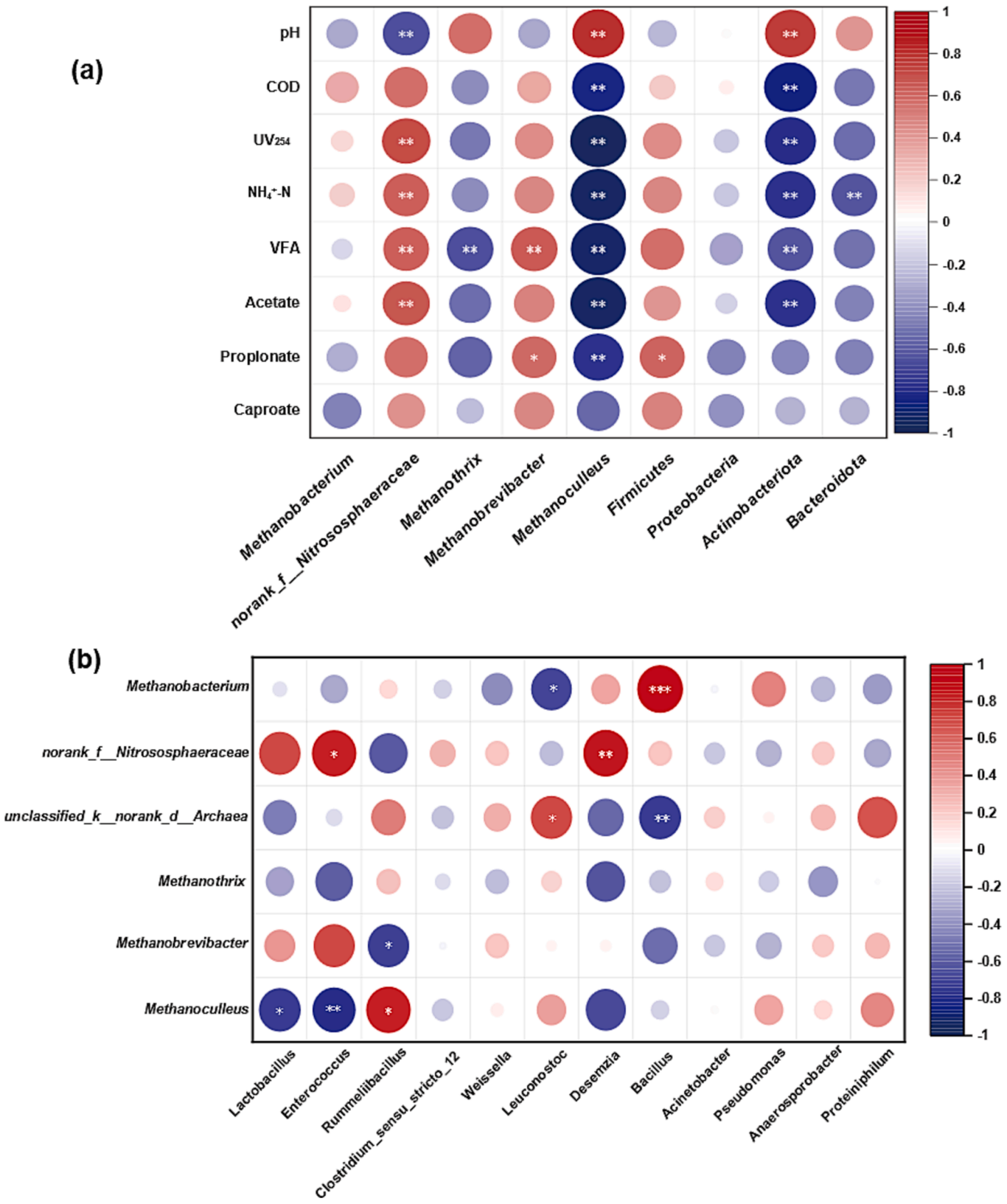


Fig. 6. (a) Correlation between environmental factors and microorganisms; (b) correlation between important bacteria and methanogens.

methanogenic bacteria indicated that the interior of this reactor is not affected by the accumulation of VFAs, which was conducive to the balance and stability of microbial community structure within the landfill.

The correlation between bacteria and archaea under micro-aeration and leachate recharge conditions (LSR-4) was analyzed as shown in Fig. 6(b). The correlation analysis of bacteria and methanogenic bacteria revealed that *Methanobacterium* was significantly positively correlated

with Bacillus, while *norank_f_Nitrososphaeraceae* was positively correlated with *Desemzia* and *Enterococcus*, as reported in a previous study. *Bacillus*, as a parthenogenic or specialized aerobic microorganism in a micro-aerated environment, was better adapted to reproduce and metabolize the substrates required by *Methanobacterium* for methanogenesis, facilitating the methanation process. In addition, the discovery of *norank_f_Nitrososphaeraceae*, as a newly discovered archaea genus level, suggested that aerobic ammonia oxidation occured in landfill piles, which facilitated the nitrification of ammonia nitrogen. *Desemzia* and *Methanobacterium*, as bacteria with high abundance in the pre-landfill period, greatly contributed to the hydrolytic acidification process of organic matter (Xing et al., 2013). The growth and reproduction of *Norank_f_Nitrososphaeraceae* was also favored. Furthermore, some negative correlations were observed. For example, *Methanoculleus* was significantly negatively correlated with *Enterococcus* and negatively correlated with *Lactobacillus*. *Unclassified_k_norank_d_Archaea* was significantly negatively correlated with *Bacillus*, and *Methanobacterium* and *Leuconostoc* were negatively correlated. This also indicated that the emergence of hydrotropic methanogenic bacteria required a certain parthenogenic and anaerobic environment, which acted as a significant negative correlation with bacteria that parthenogenic acetic acid fermentation (Chen et al., 2020). The results showed that micro-aeration and leachate recirculation into the landfill pile optimized the microbial community structure and constructed two synergistic relationships between *Methanobacterium* and *Bacillus*, *norank_f_Nitrososphaeraceae* with *Desemzia* and *Enterococcus*, which promoted the methanation process and avoided the negative effects of VFAs accumulation (Hussain et al., 2017).

3.3.4. Integrated microbial action mechanism for accelerated landfill waste stabilization

In terms of the integrated microbial action mechanism for accelerated landfill waste stabilization, micro-aeration and leachate recirculation created a unique landfill environment with overlapping anaerobic and aerobic conditions, which supported the ecological niches of anaerobic and facultative bacteria. During the early stages of landfill, the microbial community structure inside did not differ much. However, the integrated process of micro-aeration and leachate recirculation promoted a large enrichment of hydrolytic acidifying bacteria. With the increase of landfill time, micro-aeration and leachate recirculation gradually increased the abundance and diversity of the microbial community inside the landfill.

Small amounts of O_2 are introduced into anaerobic landfill environments, and cells should first use O_2 as an electron acceptor because it is thermodynamically more favorable. Microorganisms are able to adapt their own metabolic pathways from anaerobic to microaerobic, and the metabolic shift is accompanied by changes in energy acquisition. It has been shown that under micro-aeration conditions, the acetyl coenzyme associated with the synthesis of CO_2 undergoes the TCA cycle and is further oxidized to CO_2 using the electron carrier NADH, which facilitates the degradation of organic matter to generate large amounts of CO_2, allowing the entire reaction process to proceed rapidly (Zhang et al., 2021). Some studies have shown that the ATP supplied by Pyruvic acid entering the TCA cycle in a micro-oxygen environment is 8 times that produced under anaerobic conditions (Li et al., 2022a, Li et al., 2022b). The micro-aerobic environment formed by micro-aeration will enable microorganisms to undergo a complete TCA cycle, providing sufficient energy support for microbial proliferation. The dominant bacteria in LSR-4, such as *Enterococcus*, *Rummeliibacillus*, and *Clostridium_sensu_stricto_12*, which were anaerobic hydrolytic acidifiers, accelerate the microbial degradation of VFAs due to the continuous provision of micro-oxygenic environment and energy support, and provided substrates for the methanogenic phase.

Under appropriate micro-aeration conditions, methanogenic bacteria (*Methanoculleus* and *Methanobrevibacter*) gradually established oxygen tolerance under alternating micro-aerobic and anaerobic environmental conditions, and the dominant role of methanogenic bacteria in this environment became more obvious. Additionally, the continuous leachate recirculation provided water and nutrients to the landfill environment cycle, enabling microorganisms to metabolize and proliferate in a suitable environment, which promoted system stability. The integrated process of micro-aeration and leachate recirculation can also keep the balance between different microbial populations in the landfill system also between anaerobic and aerobic, between oxidation and reduction reactions, and between substrate and product.

4. Conclusion

Micro-aeration and leachate recirculation enhanced the hydrolytic acidification of microorganisms and increased the production of TVFAs by 53 %. At the end of landfill, the concentrations of COD, TP and NH_4^+–N in leachate could be reduced from the initial concentration change to 78 %, 45 % and 58 %, respectively. The organic matter was reduced from 58 % to about 25 %, which was 46 % more efficient than the degradation of conventional landfill. The species diversity and abundance of hydrolytic acidifying bacteria were increased, which could rapidly generate VFAs and produce methane from methanogenic substrates, accelerating the landfill stabilization process.

CRediT authorship contribution statement

Kaili Liu: Conceptualization, Resources, Funding acquisition, Data curation, Writing – original draft. **Longyi Lv:** Conceptualization, Investigation, Writing – original draft. **Weiguang Li:** Writing – review & editing. **Xinyuan Wang:** Supervision, Project administration, Writing – review & editing. **Muda Han:** Writing – review & editing. **Zhijun Ren:** Methodology, Writing – review & editing. **Wenfang Gao:** Supervision, Project administration, Writing – review & editing. **Pengfei Wang:** Methodology. **Xiaoyang Liu:** Methodology. **Li Sun:** Methodology. **Guangming Zhang:** Conceptualization, Investigation, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.129615.

References

Abuabdou, S.M.A., Ahmad, W., Aun, N.C., Bashir, M.J.K., 2020. A review of anaerobic membrane bioreactors (AnMBR) for the treatment of highly contaminated landfill leachate and biogas production: Effectiveness, limitations and future perspectives. J. Clean Prod. 255, 120215.

Bandala, E.R., Liu, A.n., Wijesiri, B., Zeidman, A.B., Goonetilleke, A., 2021. Emerging materials and technologies for landfill leachate treatment: A critical review. Environ. Pollut. 291, 118133.

Cao, Q., Zhang, W., Lian, T., Wang, S., Yin, F., Zhou, T., Zhang, H., Zhu, J., Dong, H., 2022. Roles of micro-aeration on enhancing volatile fatty acids and lactic acid production from agricultural wastes. Bioresour. Technol. 347, 126656.
Castro, I., Azevedo, L., Souza, C., 2023. Evaluation of microaeration strategies in the digestion zone of UASB reactors as an alternative for biogas desulfurization. Environ. Technol. 44 (4), 431–445.
Chen, W., Wang, F., Gu, Z., Li, Q., 2020. Recovery of efficient treatment performance in a semi-aerobic aged refuse biofilter when treating landfill leachate: Washing action using domestic sewage. Chemosphere 245, 125618.
Chung, J., Kim, S., Baek, S., Lee, N., Park, S., Lee, J., Lee, H., Bae, W., 2015. Acceleration of aged-landfill stabilization by combining partial nitrification and leachate recirculation: A field-scale study. J. Hazard. Mater. 285, 436–444.
Cottingham, P.D., Davies, T.H., Hart, B.T., 1999. Aeration to Promote Nitrification in Constructed Wetlands. Environ. Sci. Technol. Lett. 20 (1), 69–75.
Gardner, N., Manley, B.J.W., Pearson, J.M., 1993. Gas emissions from landfills and their contributions to global warming. Appl. Energy 44 (2), 165–174.
Gou, C., Yang, Z., Huang, J., Wang, H., Xu, H., Wang, L., 2014. Effects of temperature and organic loading rate on the performance and microbial community of anaerobic co-digestion of waste activated sludge and food waste. Chemosphere 105, 146–151.
Grossule, V., Lavagnolo, M.C., 2020. Lab tests on semi-aerobic landfilling of MSW under varying conditions of water availability and putrescible waste content. J. Environ. Manage. 256, 109995.
Gu, N., Liu, J., Ye, J., Chang, N., Li, Y.-Y., 2019. Bioenergy, ammonia and humic substances recovery from municipal solid waste leachate: A review and process integration. Bioresour. Technol. 293, 122159.
Hamid, H., Li, L.Y., Grace, J.R., 2020. Effect of substrate concentrations on aerobic biotransformation of 6:2 fluorotelomer sulfonate (6:2 FTS) in landfill leachate. Chemosphere 261, 128108.
Hettiaratchi, J.P.A., Jayasinghe, P.A., Bartholameuz, E.M., Kumar, S., 2014. Waste degradation and gas production with enzymatic enhancement in anaerobic and aerobic landfill bioreactors. Bioresour. Technol. 159, 433–436.
Hossain, M.S., Penmethsa, K.K., Hoyos, L., 2009. Permeability of municipal solid waste in bioreactor landfill with degradation. Geotech. Geol. Eng. 27 (1), 43–51.
Hu, Y.S., Zhao, Y.Q., Zhao, X.H., Kumar, J.L.G., 2012. High Rate Nitrogen Removal in an Alum Sludge-Based Intermittent Aeration Constructed Wetland. Environ. Sci. Technol. 46 (8), 4583–4590.
Huang, D., Du, Y., Xu, Q., Ko, J.H., 2022. Quantification and control of gaseous emissions from solid waste landfill surfaces. J. Environ. Manage. 302, 114001.
Hussain, A., Filiatrault, M., Guiot, S.R., 2017. Acidogenic digestion of food waste in a thermophilic leach bed reactor: Effect of pH and leachate recirculation rate on hydrolysis and volatile fatty acid production. Bioresour. Technol. 245, 1–9.
Ike, N.N.S.A., 2009. Effects of intermittent and continuous aeration on accelerative stabilization and microbial population dynamics in landfill bioreactors. J. Biosci. Bioeng. 108 (4), 336–343.
Ishigaki, T., Sugano, W., Nakanishi, A., Tateda, M., Ike, M., Fujita, M., 2004. The degradability of biodegradable plastics in aerobic and anaerobic waste landfill model reactors. Chemosphere 54 (3), 225–233.
Kim, M.S., Kim, D.H., Yun, Y.M., 2017. Effect of operation temperature on anaerobic digestion of food waste: Performance and microbial analysis. Fuel 209, 598–605.
Lay, J.J., Li, Y.Y., Noike, T., 2014. Developments of bacterial population and methanogenic activity in a laboratory-scale landfill bioreactor. Water Res. 32 (12), 3673–3679.
Leckie, J.O., Pacey, J.G., Halvadakis, C., 1979. Landfill management with moisture control. J. Environ. Eng. Div. 105 (2), 337–355.
Lee, H., Coulon, F., Beriro, D.J., Wagland, S.T., 2022. Recovering metal(loids) and rare earth elements from closed landfill sites without excavation: Leachate recirculation opportunities and challenges. Chemosphere 292, 133418.
Li, W., Liu, Y., Wu, B., Gu, L.i., Deng, R., 2022a. Upgrade the high-load anaerobic digestion and relieve acid stress through the strategy of side-stream micro-aeration: biochemical performances, microbial response and intrinsic mechanisms. Water Res. 221, 118850.
Li, X., Su, M., Chen, Y., Nisa, M.U., Zhao, N., Jiang, X., Li, Z., 2022b. The modification of titanium in mesoporous silica for Co-based Fischer-Tropsch catalysts. Front. Chem. Sci. Eng. 16 (8), 1224–1236.
Liu, K., Lv, L., Li, W., Ren, Z., Wang, P., Liu, X., Gao, W., Sun, L., Zhang, G., 2023. A comprehensive review on food waste anaerobic co-digestion: Research progress and tendencies. Sci. Total Environ., 163155
Luo, L., Wong, J.W.C., 2019. Enhanced food waste degradation in integrated two-phase anaerobic digestion: Effect of leachate recirculation ratio. Bioresour. Technol. 291, 121813.
Lv, Y., Chang, N., Li, Y.-Y., Liu, J., 2021. Anaerobic co-digestion of food waste with municipal solid waste leachate: A review and prospective application with more benefits. Resour. Conserv. Recycl. 174, 105832.
Matsuto, T., Zhang, X., Matsuo, T., Yamada, S., 2015. Onsite survey on the mechanism of passive aeration and air flow path in a semi-aerobic landfill. Waste Manage. 36, 204–212.
NBSC (National Bureau of Statistics of China), 2021. China Statistical Yearbook 2021.
Nghiem, L.D., Koch, K., Bolzonella, D., Drewes, J.E., 2017. Full scale co-digestion of wastewater sludge and food waste: Bottlenecks and possibilities. Renew. Sust. Energ. Rev. 72, 354–362.
Nguyen, D., Wu, Z., Shrestha, S., Lee, P.-H., Raskin, L., Khanal, S.K., 2019. Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res. 166, 115080.
Ouellet-Plamondon, C., Chazarenc, F., Comeau, Y., Brisson, J., 2006. Artificial aeration to increase pollutant removal efficiency of constructed wetlands in cold climate. Ecol. Eng. 27 (3), 258–264.
Peng, Y., Li, L., Yuan, W., Wu, D.i., Yang, P., Peng, X., 2022. Long-term evaluation of the anaerobic co-digestion of food waste and landfill leachate to alleviate ammonia inhibition. Energy Conv. Manag. 270, 116195.
Pinjing, H.N.Y.H., 2011. N2O and NH3 emissions from a bioreactor landfill operated under limited aerobic degradation conditions. J. Environmental. Sci. 23 (6), 1011–1019.
Prantl, R., Tesar, M., Huber-Humer, M., Lechner, P., 2006. Changes in carbon and nitrogen pool during in-situ aeration of old landfills under varying conditions. Waste Manage. 26 (4), 373–380.
Renou, S., Givaudan, J.G., Poulain, S., Dirassouyan, F., Moulin, P., 2008. Landfill leachate treatment: Review and opportunity. J. Hazard. Mater. 150 (3), 468–493.
Shang, Z., Wang, R., Zhang, X., Tu, Y., Sheng, C., Yuan, H., Wen, L., Li, Y., Zhang, J., Wang, X., Yang, G., Feng, Y., Ren, G., 2023. Differential effects of petroleum-based and bio-based microplastics on anaerobic digestion: A review. Sci. Total Environ. 875, 162674.
Stabnikova, O., Liu, X., Wang, J., 2008. Anaerobic digestion of food waste in a hybrid anaerobic solid–liquid system with leachate recirculation in an acidogenic reactor. Biochem. Eng. J. 41 (2), 198–201.
Tong, H., Yin, K., Giannis, A., Ge, L., Wang, J., 2015. Influence of temperature on carbon and nitrogen dynamics during in situ aeration of aged waste in simulated landfill bioreactors. Bioresour. Technol. 192, 149–156.
Vavilin, V.A., Rytov, S.V., Lokshina, L.Y., Pavlostathis, S.G., Barlaz, M.A., 2010. A distributed model of solid waste anaerobic digestion: sensitivity analysis. J. Biosci. Bioeng. 81 (1), 66–73.
Viegas, C., Nobre, C., Mota, A., Vilarinho, C., Gouveia, L., Gonçalves, M., 2021. A circular approach for landfill leachate treatment: Chemical precipitation with biomass ash followed by bioremediation through microalgae. J. Environ. Chem. Eng. 9 (3), 105187.
Vigil, M., Marey-Pérez, M.F., Martinez Huerta, G., Álvarez Cabal, V., 2015. Is phytoremediation without biomass valorization sustainable? — Comparative LCA of landfilling vs. anaerobic co-digestion. Sci. Total Environ. 505, 844–850.
Wang, S., Hu, Z.-Y., Geng, Z.-Q., Tian, Y.-C., Ji, W.-X., Li, W.-T., Dai, K., Zeng, R.J., Zhang, F., 2022. Elucidating the production and inhibition of melanoidins products on anaerobic digestion after thermal-alkaline pretreatment. J. Hazard. Mater. 424, 127377.
Wang, Y.-N., Xu, R., Wang, H., Shi, H., Kai, Y., Sun, Y., Li, W., Bian, R., Zhan, M., 2021. Insights into the stabilization of landfill by assessing the diversity and dynamic succession of bacterial community and its associated bio-metabolic process. Sci. Total Environ. 768, 145466.
Wu, C., Shimaoka, T., Nakayama, H., Komiya, T., Chai, X., Hao, Y., 2014. Influence of aeration modes on leachate characteristic of landfills that adopt the aerobic-anaerobic landfill method. Waste Manage. 34 (1), 101–111.
Wu, C., Shimaoka, T., Nakayama, H., Komiya, T., Chai, X., 2016. Stimulation of waste decomposition in an old landfill by air injection. Bioresour. Technol. 222, 66–74.
Xing, W., Lu, W., Zhao, Y., Zhang, X., Deng, W., Christensen, T.H., 2013. Environmental impact assessment of leachate recirculation in landfill of municipal solid waste by comparing with evaporation and discharge (EASEWASTE). Waste Manage. 33 (2), 382–389.
Xu, S., Zhang, H., 2022. First evidence for anaerobic oxidation of methane process in landfill cover soils: Activity and responsible microorganisms. Sci. Total Environ. 841, 156790.
Yang, Y., Yao, X., Wu, S., Wang, X., Feng, L., Feng, X., Ma, Y., 2023. Enhanced treatment of azo dyes in wastewater using heat-activated persulfate with micro-nano bubble aeration. Chem. Eng. Res. Des. 197, 24–37.
Yu, N., Guo, B., Zhang, Y., Zhang, L., Zhou, Y., Liu, Y., 2020. Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater. Water Res. 177, 115783.
Yu, F., Wu, Z., Wang, J., Li, Y., Chu, R., Pei, Y., Ma, J., 2022. Effect of landfill age on the physical and chemical characteristics of waste plastics/microplastics in a waste landfill sites. Environ. Pollut. 306, 119366.
Zhang, H., Chen, F., Xu, J., Zhang, J., Han, Y., 2022a. Chemical reactions of oily sludge catalyzed by iron oxide under supercritical water gasification condition. Front. Chem. Sci. Eng. 16 (6), 886–896.
Zhang, G., Liu, K., Lv, L., Gao, W., Li, W., Ren, Z., Yan, W., Wang, P., Liu, X., Sun, L.i., 2023. Enhanced landfill process based on leachate recirculation and micro-aeration: A comprehensive technical, environmental, and economic assessment. Sci. Total Environ. 857, 159535.
Zhang, Z., Yu, Y., Xi, H., Zhou, Y., 2021. Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater. J. Clean Prod. 317, 128343.
Zhang, Z., Xi, H., Yu, Y., Wu, C., Yang, Y., Guo, Z., Zhou, Y., 2022b. Coupling of membrane-based bubbleless micro-aeration for 2,4-dinitrophenol degradation in a hydrolysis acidification reactor. Water Res. 212, 118119.
Zheng, Z., Li, W., Zhang, D., Qin, W., Zhao, Y.i., Lv, L., 2019. Effect of iron and manganese on ammonium removal from micro-polluted source water by immobilized HITLi7T at 2 °C. Bioresour. Technol. 285, 121367.
Zhu, H., Parker, W., Conidi, D., Basnar, R., Seto, P., 2011. Eliminating methanogenic activity in hydrogen reactor to improve biogas production in a two-stage anaerobic digestion process co-digesting municipal food waste and sewage sludge. Bioresour. Technol. 102 (14), 7086–7092.